globalServiceDiscovery

Service Description

Service ID: *"globalServiceDiscovery"*

Abstract
This document describes the `globalServiceDiscovery` service, version 5.2.2, produced by the Gatekeeper support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `globalServiceDiscovery` service, which provides discovery of authorized providers of a service in the neighbour clouds, through the inter-cloud relay. It is produced by the Gatekeeper system [1] in service version 5.2.2.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

Successor of the Arrowhead 4.6 Gatekeeper global service discovery (GSD); compatible with its paths (`/init_gsd`).

2.2 How This Service Is Meant to Be Used

DynamicServiceOrchestration calls `query` during inter-cloud orchestration with the service requirement and the consumer. The Gatekeeper asks the Gatekeepers of the target clouds over the relay; each answers only with the providers its ConsumerAuthorization grants to the consumer of this cloud.

2.3 Important Delimitations and Dependencies

Only configured peer clouds (`FEDERATION_PEER_ORGANIZATIONS`) are asked; the answering cloud's identity must match. The profile is `legacy-mtls`: TLS per relay connection, not end-to-end encryption.

3 Service Interface

This section describes the interfaces to the `globalServiceDiscovery` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.



Figure 1: SysML-style block of the `globalServiceDiscovery` service and its operations.

The following interface operations are available.

3.1 operation query (GlobalServiceDiscoveryRequest) : GlobalServiceDiscoveryResult

Asks the target clouds for authorized providers of the required service.

3.2 operation multi-query (GlobalServiceDiscoveryRequest) : GlobalServiceDiscoveryResult

Same as `query` (v4 multi-GSD path).

4 Information Model

Here, all data objects that can be part of the `globalServiceDiscovery` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.4, which are used to represent things like hashes and identifiers.

4.1 struct GlobalServiceDiscoveryRequest

Field	Type	Description
serviceRequirement	Map<String, Object>	Orchestration service requirement; `serviceDefinition` is required, `operations` are checked (alias `requestedService`).
targetClouds	List<String>	Clouds to ask (`<Cloud>` or `<Cloud>|<Organization>`); default all configured peer clouds.
consumerSystem	String	Consumer on whose behalf the orchestrator asks; default the requester.

4.2 struct GlobalServiceDiscoveryResult

Field	Type	Description
results	List<InterCloudProvider>	Authorized providers of all asked clouds.
unsuccessfulRequests	Number	Number of clouds that did not answer correctly.
queriedClouds	Number	Number of clouds asked.

4.3 struct InterCloudProvider

Field	Type	Description
cloud	Name	Provider cloud, `<Cloud>|<Organization>`.
providerName	Name	Provider system.
serviceDefinition	Name	Service definition.
version	Version	Service version.
metadata	Metadata	Instance metadata.
operations	List<String>	Offered operations.

4.4 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.
Version	Specifies a version, semantic versioning `major.minor.patch`.

5 References

[1] J. Delsing, “Gatekeeper – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-Gatekeeper-java-spring/documentation/v5.3.0/Gatekeeper-v5.3.0-SysD.pdf.

[2] ——, “globalServiceDiscovery HTTP/TLS/JSON – Interface Design Description (Gatekeeper),” Eclipse Arrowhead, IDD 5.3.0, 2026, gatekeeper-v5.3.0-globalServiceDiscovery-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	